Uppsala Monitoring Centre

March 2026

# WHODrug User Guide

**Applicable for the B3 and C3 formats**

*UMC's Drug Dictionary – WHODrug Global is a global and standardised terminology for medicines and vaccines, developed, maintained and distributed by Uppsala Monitoring Centre (UMC), an independent, not-for-profit foundation.*

# 1 Content

1 Content ..... 1
2 Disclaimer ..... 3
3 Changes from previous version ..... 3
4 Getting started ..... 4
4.1 Introduction ..... 4
4.2 The WHODrug Global product portfolio ..... 4
4.2.1 The WHODrug terminology ..... 4
4.2.2 WHODrug Standardised Drug Groupings (SDG) ..... 4
4.2.3 WHODrug Koda ..... 4
4.2.4 WHODrug Cross Reference ATC 5 ..... 5
4.2.5 WHODrug Cross Reference Tool (CRT) Japan ..... 5
4.2.6 WHODrug Link Korea ..... 5
4.3 Additional resources ..... 5
5 WHODrug Global ..... 7
5.1 The terminology of WHODrug ..... 7
5.2 Terminology formats ..... 7
5.2.1 The B3 and the C3 formats ..... 7
5.2.2 The B2 and the C formats ..... 8
6 Core WHODrug concepts ..... 9
6.1 Drug name ..... 9
6.1.1 Product name ..... 9
6.1.2 Active substance name ..... 9
6.1.3 Umbrella records ..... 11
6.1.4 NOS substances ..... 12
6.1.5 Non-unique names ..... 12
6.2 Drug code ..... 13
6.2.1 Drug Record Number ..... 13
6.2.2 Sequence number 1 ..... 14
6.2.3 Sequence number 2 ..... 14
6.3 C3 format specific concepts ..... 15
6.3.1 Record Identifier (RID) ..... 15
6.3.2 Name specifier ..... 16
6.3.3 Archived record ..... 16
7 The terminology of WHODrug Global Chinese ..... 18
7.1 Drug information ..... 18
7.2 Pinyin names ..... 19
7.3 Non-unique names ..... 19
7.4 Umbrella records ..... 20
8 ISO IDMP data and mappings to WHODrug ..... 21
8.1 Pharmaceutical Product ID (PhPID) ..... 21
8.2 Global Substance Identifiers (GSIDs) ..... 22
8.3 Strength ..... 22
8.4 Dose Form ..... 22
8.4.1 Basic dose form ..... 22
8.4.2 Administration method ..... 22
8.4.3 Intended site ..... 22



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Box 1051, SE-751 40 Uppsala, Sweden
Email: support@who-umc.org, www.who-umc.org

8.4.4 Release characteristics ..... 22
**9 Accessing WHODrug data ..... 23**
**9.1 WHODrug Insight ..... 23**
**9.2 WHODrug Global file packages ..... 23**
9.2.1 WHODrug Global standard files ..... 23
9.2.2 WHODrug Global Chinese standard files ..... 24
9.2.3 ISO IDMP data files and mapping to WHODrug ..... 24
9.2.4 WHODrug Global additional features files ..... 24
9.2.5 WHODrug Global Chinese additional feature files ..... 25
9.2.6 WHODrug Standardised Drug Groupings files ..... 25
**10 ATC classification ..... 26**
**10.1 WHO Collaborating Centre for Drug Statistics Methodology ATC classification system ..... 26**
**10.2 Herbal ATC ..... 27**
**10.3 ATC assignments in WHODrug ..... 27**
10.3.1 Official and UMC assigned ATC codes ..... 28
10.3.2 UMC created ATC codes ..... 28
10.3.3 Inherited ATC assignments in WHODrug ..... 29
10.3.4 ATC assignments in the B3 format ..... 30
10.3.5 ATC assignments in the C3 format ..... 31
**11 Changes to the WHODrug data ..... 32**
**11.1 WHODrug Change Analysis Tool ..... 32**
**11.2 WHODrug Change Request ..... 32**
**Appendix 1 ..... 34**



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Email: support@who-umc.org, www.who-umc.org

# 2 Disclaimer

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If you do not recognise a feature that is described in this guide it may be due to the set-up of your coding or analysis software or the area in which you use WHODrug. You are always welcome to contact UMC at support@who-umc.org or your software vendor if you have any questions.

# 3 Changes from previous version

| Version | Summary of changes |
|---|---|
| **March 2026** | Numbered sections added.<br>Core WHODrug concepts: Major rearrangement of the text. Image added. Example numbers 2-12 changed.<br>Core WHODrug concepts: “Preferred name” changed to “substance name”.<br>Archived record: concept name changed from “Old form” to “Archived record”.<br>ISO IDMP data and mappings to WHODrug: New chapter added<br>Accessing WHODrug data: New main heading. Insight section moved further up.<br>Accessing WHODrug data: “ISO IDMP data files and mapping to WHODrug” and “Archive record” file added, txt files deleted |
| **September 2025** | No changes. |
| **March 2025** | No changes. |
| **September 2024** | Old form: information changed and removed, example changed, and chapter moved. |



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Box 1051, SE-751 40 Uppsala, Sweden
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# 4 Getting started

## 4.1 Introduction

The UMC Drug Dictionary WHODrug product portfolio consists of a comprehensive drug reference terminology and related products. WHODrug contains information about worldwide medicinal products intended for human use, their active substances, and relevant classifications and groupings.

WHODrug can be used to structure and analyse clinical trial and post-marketing safety data and is used by pharmaceutical companies, CROs, research organisations and regulatory agencies, in the fields of clinical trials and pharmacovigilance.

Since March 1, 2017, all WHODrug terminology types are combined into one terminology named WHODrug Global. In September 1, 2019, a Chinese version of WHODrug Global, WHODrug Global Chinese, was introduced to enable drug coding in Chinese.

To keep WHODrug current and comprehensive, the terminology is updated on a regular basis and updated versions released twice per year.

**The WHODrug User Guide**

This user guide aims to give theoretical background information and describe the features of WHODrug. This guide can help you to get started if you are new to WHODrug, but it can also be used as a look-up document if you are an experienced user. The WHODrug User Guide is available in English, Chinese, Spanish and Portuguese.

## 4.2 The WHODrug Global product portfolio

### 4.2.1 The WHODrug terminology

WHODrug is a consistent and structured drug terminology with exact terminology, containing information about medicinal products worldwide. The terminology is used to identify drug names and evaluate medicinal product information. All data are available in a data file package and in the web-based browser WHODrug Insight.

### 4.2.2 WHODrug Standardised Drug Groupings (SDG)

WHODrug Standardised Drug Groupings (SDGs) are standardised and unbiased groupings of drugs, that can identify drugs with the same properties. The SDGs can be used for example in various types of safety analysis, or in the specification of exclusion criteria in clinical trials. All SDGs allow for modifications by the user, and when altered they are referred to as Customised Drug Groupings (CDGs). The SDGs are included in the WHODrug Global files package and in the web-based browser WHODrug Insight. Read more about the SDGs in the WHODrug Standardised Drug Groupings User Guide.

### 4.2.3 WHODrug Koda

WHODrug Koda is a fully automated drug coding assistant that combines the use of artificial intelligence by machine learning, thorough and continual training of the engine, industry best practices and UMC's accumulated insights on drug coding. The service is specifically designed to increase the efficiency, consistency, and quality of drug coding, with the end goal of safer use of medicines. The service is provided as an API, as well as an intuitive web application. Read more about Koda in the WHODrug Koda User guide.



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#### 4.2.4 WHODrug Cross Reference ATC 5

WHODrug Cross Reference ATC 5 (CR ATC 5) provides a connection between the available 4th level ATCs in WHODrug and the 5th level ATCs based on the active ingredient given. The CR ATC 5 mapping table enables efficient assignment of the ATC 5th level codes for XEVMPD submission to comply with European Medicines Agency (EMA) requirements. WHODrug CR ATC 5 is available as an add-on to an existing WHODrug Global subscription. Read more about the CR ATC 5 in the WHODrug Cross Reference ATC 5 User Guide.

#### 4.2.5 WHODrug Cross Reference Tool (CRT) Japan

WHODrug Cross Reference Tool Japan (CRT Japan) converts Japanese Iyakuhinmei Data File (IDF) codes into WHODrug codes. To comply with Japanese Pharmaceuticals and Medical Devices Agency (PMDA) recommendations to use WHODrug for New Drug Applications, CRT Japan eliminates time-consuming double-coding by directly matching domestic IDF codes and international WHODrug codes. Read more about CRT Japan in the WHODrug Cross Reference Tool Japan User Guide.

#### 4.2.6 WHODrug Link Korea

WHODrug Link Korea is a bridge between WHODrug codes and local Korean codes, by providing mapping files that convert WHODrug codes for Korean medications to the required Korean drug codes. WHODrug Link Korea facilitates E2B(R3) compliance for ICSR submissions to the Republic of Korea's Ministry of Food and Drug Safety (MFDS), and efficiently reduces the time needed for manual code selection. Read more about Link Korea in the WHODrug Link Korea User Guide.

### 4.3 Additional resources

Besides this user guide, as mentioned below there are several other guides, as well as documents, communication channels and events covering different areas and applications of WHODrug. All available documentation (described in more detail overleaf) can be found in the WHODrug section of the UMC website.

**How can I get access to WHODrug resources?**

All users of WHODrug currently working for an organisation with a valid license are eligible for a WHODrug account. Apply for your personal account from the WHODrug Access manager in your organisation, or by contacting WHODrug customer support at Support@who-umc.org.



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### *Core WHODrug documentation*

| | |
|---|---|
| WHODrug Best Practices | A document developed together with coding experts in the WHODrug user community, offering descriptions of best practice procedures in various areas. Available in English and Chinese. |
| How to use the WHODrug C3 format for drug coding | This document offers best practices when coding drug information with WHODrug Global C3 format. It is written by UMC staff and does not stipulate any regulatory requirements. Provided in English,Spanish and Portuguese. |

### *Guides to additional WHODrug products*

| | |
|---|---|
| WHODrug Cross Reference ATC 5 User Guide | The Cross Reference ATC 5 User guide includes a product description and file description of the cross-reference table. |
| WHODrug Cross Reference Tool Japan User Guide | This user guide describes how Cross Reference Tool (CRT) Japan enables users to specify the WHODrug Drug Code corresponding to a selected code in the Iyakuhinmei Data File (IDF). Available in English and Japanese. |
| WHODrug Koda User Guide | The WHODrug Koda User Guide includes a description of the drug name and ATC coding process, and how to review the result when using the automated coding engine WHODrug Koda. |
| WHODrug Link Korea User Guide | This user guide provides guidance on how to make the connection between WHODrug Drug codes and local Korean drug codes. Available in English and Korean. |
| WHODrug Standardised Drug Groupings User Guide | This document includes an introduction to the Standardised Drug Groupings (SDGs), as well as detailed information about all individual SDGs. It is released twice a year in English and Chinese. |

### *Continuously updated WHODrug documentation*

| | |
|---|---|
| WHODrug Version verification | This document is intended for verification of the version and format (before 2021 also type) of WHODrug that has been implemented. The WHODrug Version verification document is updated biannually, alongside the releases of WHODrug. |
| What's New in WHODrug | This annual document outlines the most important new developments and improvements in WHODrug and is published with the March 1 release of WHODrug. |
| WHODrug Newsletter | The WHODrug Newsletter is published on a quarterly basis and highlights relevant new developments and improvements in WHODrug. |

### *Other UMC resources available for WHODrug users*

| | |
|---|---|
| WHODrug webinars | All WHODrug users with an account are invited to attend webinars covering different WHODrug related topics. The webinars are in general held once a month on Tuesdays, at 9 am CE(S)T and 5 pm CE(S)T. |
| WHODrug support | All WHODrug users have access to exclusive WHODrug support via e-mail (support@who-umc.org). |
| WHODrug User Group meetings | Each year a number of WHODrug User Group meetings are held around the world. Join these meetings to learn more about WHODrug and interact and learn from other WHODrug users. |
| Introduction to WHODrug | This self-paced online course provides basic knowledge about the WHODrug content and structure, as well as basic coding concepts necessary to standardize data using WHODrug. Available in English, Spanish and Portuguese. |



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Box 1051, SE-751 40 Uppsala, Sweden
Email: support@who-umc.org, www.who-umc.org

# 5 WHODrug Global

## 5.1 The terminology of WHODrug

Uppsala Monitoring Centre (UMC) provides the most comprehensive and actively used drug reference terminology in the world. The terminology of WHODrug is used for structuring clinical trial and post-marketing safety data. It contains medicinal products and active substances intended for human and medicinal use, both conventional and of natural origin. Examples of the types of medicinal products/active substances included in WHODrug are small synthetic or semi-synthetic chemical substances, biotherapeutics, herbal remedies (including fungi and algae), vaccines, dietary supplements, radio pharmaceuticals, diagnostic agents, traditional medicine (such as Traditional Chinese Medicine (TCM) and Kampo), and homeopathic remedies. The terminology coverage includes prescription-only products, over-the-counter (OTC) and pharmacist-dispensed preparations. In addition, investigational active substances may also be included in WHODrug.

There are data elements in WHODrug representing product names and active substance names. In addition, there are data elements representing classes of drugs for the coding of imprecise verbatim terms, i.e., Umbrella records and Not Otherwise Specified substances (NOS substances). To summarise, the following types of data records are available in WHODrug:

- Product names
- Active substances (bases and variations thereof)
- Records representing defined groups of drugs (i.e. Umbrella records and Not Otherwise Specified substances (NOS substances)

UMC collaborates with national pharmacovigilance centres as well as several institutions and organizations around the world to maintain the terminology content up-to-date. Examples of these institutions are The Royal Botanic Gardens Kew, WHO Collaborating Centre for Drug Statistics Methodology, and IQVIA, an organisation which collects global drug information in over 100 countries. In addition, drug names mentioned in individual case safety reports (ICSRs) sent to UMC are included, as well as products and substances registered by the U.S Food and Drug Administration (FDA) and the European Medicines Agency (EMA). User requested Change Requests are also a valuable source of data.

The drug information is verified in at least one reliable reference before being included in WHODrug. The references used to verify the medicinal products are either official national drug information databases or other reliable national and international drug reference sources.

WHODrug contains data from 1968 onwards. Even if a drug has been withdrawn from the market it is not removed from WHODrug. The reason for this is that the information might be used to code old data.

## 5.2 Terminology formats

### 5.2.1 The B3 and the C3 formats

The terminology of WHODrug is currently presented in two formats, the B3 format and the C3 format. The difference between these two is the amount of information displayed for each drug name, i.e. the level of detail in which the record is described. The data presented in the two formats are differentiated by table structures and data fields. All drug names available in the B3 format are also available in the C3 format and vice versa. Both formats are made available for all users of WHODrug.



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The B3 format contains information about drug names, active substances and ATC classifications.

As well as containing all the B3 format information, the C3 format includes additional data elements, namely information about the country of sale, marketing authorisation holder, pharmaceutical form according to the EphMRA NFC classification system, and strength. The C3 format also has a higher precision of ATC code assignments since each drug name is assigned specific ATC codes. The C3 format can either be used as a reference for additional information while performing coding in the B3 format, or as a stand-alone coding terminology.

**B3**

| Drug code | Drug name | Active substance(s) | ATC |
|---|---|---|---|
| 00764801074 | Moveril | Thiocolchico-side | M03BX, Other centrally acting agents *official* |

**C3**

| Drug code | Drug name | Active substance(s) | ATC | Country | Marketing authorisation holder | Pharmaceutical form | Strength | Record ID |
|---|---|---|---|---|---|---|---|---|
| 00764801074 | Moveril | Thiocolchico-side | M03BX, Other centrally acting agents *official* | Italy | Acarpia services farm | Ampoules, intramuscular | 4 mg | 1931652 |

**Example 1.** The record Moveril in the B3 format and in the C3 format in WHODrug.

### 5.2.2 The B2 and the C formats

In September 2018 the files of the old B2/C formats of WHODrug were released for the last time. Since March 2019, the terminology is only provided in the new B3/C3 formats.

For WHODrug guidance on the B2 and C formats, please refer to version 2.0 of this document.



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Box 1051, SE-751 40 Uppsala, Sweden
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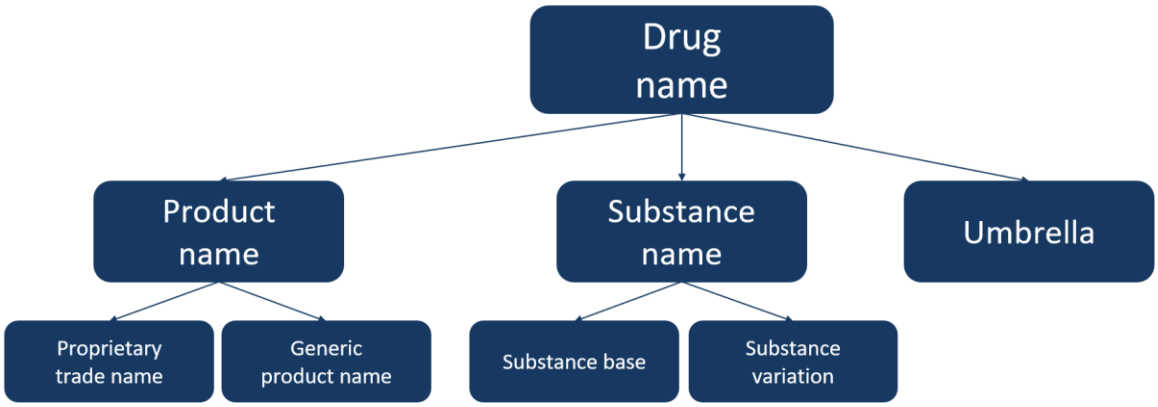
# 6 Core WHODrug concepts

## 6.1 Drug name

In WHODrug the drug name can be both a proprietary product name, such as a trade name, or a non-proprietary generic product name. The drug name can also represent a substance, substance combination, or an imprecise verbatim (Umbrella).




### 6.1.1 Product name

In WHODrug the product name represents the name of a medicinal product and can be a trade name or a generic product name.

WHODrug is a global terminology containing drug information from all around the world. The characters or letters in the name of a medicinal product might differ in the various countries where it is marketed. In WHODrug it is key to harmonise these names to enable consistent coding. This harmonisation includes how special characters are used in WHODrug. For example, hyphens (-) are not included in the product name, while other special characters, such as +, / and &, are included, following the prevailing grammatical rules.

Trademark, registered trademark, copyright symbols etc. are never included in trade names in WHODrug, to enable drug name harmonisation at a global level.

**Please note**

Some product names in WHODrug may not follow the general naming convention used in WHODrug for historical reasons or for reasons of harmonisation.

### 6.1.2 Active substance name

An active substance in WHODrug is a substance responsible for the physiological and/or pharmacological effect of a medicine. An active substance can be, for example, a chemical substance, an herbal plant or plant part, or a biological substance, and is defined by a globally recognised non-proprietary name or with chemical or botanical nomenclature as standard. Active substances are used as ingredients for medicinal products in WHODrug.

The substance name is generally an International Nonproprietary Name (INN), but if no INN exists, another, globally recognised nonproprietary name (e.g., a United States Adopted Name



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(USAN), a Japanese Accepted Name (JAN) or a chemical name) is used. For herbal records, the accepted scientific name is the substance name.

Sometimes an investigational active substance is given a company specific name called company code before receiving an official substance name. These company codes are included in WHODrug as regular substances.

To allow for flexible analysis of the coded data, substance names are available on two levels for records in WHODrug; the active substance base and the active substance variation.

#### 6.1.2.1 Active substance base

For conventional (i.e. non-herbal) remedies, the substance base refers to the specific chemical or biological active entity, for herbal remedies the substance base refers to the scientific plant name, for insulins it refers to the origin of the insulin or the insulin analogue (e.g. insulin human and insulin lispro), and for vaccine substances, the substance base refers to a vaccine for a disease/pathogen (e.g. polio vaccine).

#### 6.1.2.2 Active substance variation

Active substance variations represent different forms or alterations of the active substance.

For conventional (i.e. non-herbal) remedies, the substance variation refers to different variants for example salts and esters, for herbal remedies the substance variation refers to plant parts, for insulins, substance variation refers to various insulin derivates (e.g. isophane and crystalline insulins), and for vaccine substances, substance variation refers to various degrees of granularity (e.g. Polio vaccine live oral type 1 and Polio vaccine inact).

| Substance(s) | Substance base(s) |
|---|---|
| Ampicillin | Ampicillin |
| Ampicillin sodium | Ampicillin |
| Ampicillin;Cloxacillin | Ampicillin;Cloxacillin |
| Ampicillin sodium;Cloxacillin sodium | Ampicillin;Cloxacillin |

**Example 2**. Substance base names and substance variation names.

#### 6.1.2.3 Substance synonym

In WHODrug, an active substance is preferably an INN. In addition, there can be several other generic name records representing a substance, with names based on other name standards than INN. These are called substance synonym names.

| Drug name | Type of name |
|---|---|
| Paracetamol | Substance name (INN) |
| Acetaminophen | Substance synonym name |
| Dymadon | Product name |
| Paracetamol accord | Product name |
| Duocid | Product name |
| Ampicillin;Sulbactam | Substance name (INN) |

**Example 3**. Examples of substance, substance synonym and product names.



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**Generic name**

All active substance bases, variations and synonyms have generic names in WHODrug. Medicinal products can also have generic product names.

#### 6.1.2.4 Spanish and Portuguese translations

To enhance the usability and accessibility of WHODrug for users in Spanish- and Portuguese-speaking regions, selected active substance names are available with Spanish and/or Portuguese translation. These translated names share the same Drug code as the English substance name.

### 6.1.3 Umbrella records

An Umbrella record is a record representing a drug category or a placeholder record rather than a specific drug name. Two examples of Umbrella records are Corticosteroids and Placebo. Umbrella records are used for coding imprecise verbatim terms when coding to specific drug names is not possible. Umbrella records are identified by having a Drug code starting with 9, and they always lack ingredients. There are two types of Umbrella records, ATC Umbrella records and UMC created Umbrella records.

#### 6.1.3.1 ATC Umbrella records

ATC codes from the WHO Collaborating Centre for Drug Statistics Methodology in Oslo, Norway (WHO CC Oslo) and UMC Herbal ATC (HATC) codes are used to classify records in WHODrug. These different ATC codes are also available as ATC Umbrella records in WHODrug. ATC Umbrella records generally have names corresponding to the ATC code texts. Some ATC texts have been altered for consistency. The few ATC texts that are equivalent to Generic records in WHODrug are not represented by ATC Umbrella records.

Read more about the ATC classification system in the chapter “ATC classification”.

| ATC code | ATC text | Drug code | Umbrella record name |
|---|---|---|---|
| A | Alimentary tract and metabolism | 90061601001 | Alimentary tract and metabolism |
| A01 | Stomatological preparations | 90114401001 | Stomatological preparations |
| A01A | Stomatological preparations | 90114401001 | Stomatological preparations |
| A01AA | Caries prophylactic agents | 90059101001 | Caries prophylactic agents |
| C | Cardiovascular system | 90108501001 | Cardiovascular system |
| C02 | Antihypertensives | 90011901001 | Antihypertensives |
| C02A | Antiadrenergic agents, centrally acting | 90012001001 | Antiadrenergic agents, centrally acting |
| C02AB | Methyldopa | 00000101001 | Methyldopa |

**Example 4**. ATC Umbrella records and the substance name Methyldopa.



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#### 6.1.3.2 UMC created Umbrella records

In addition to the ATC Umbrella records there are other Umbrella records created by UMC. Most of these additional Umbrella records have been requested by WHODrug users and often represent commonly reported verbatim terms. UMC-created Umbrella records can either represent classes of drugs, or be broad placeholder records.

| Drug name | Drug code |
|---|---|
| Placebo | 90046801001 |
| Drug not administered | 90128901001 |
| Contraceptives | 90113401001 |
| Artificial tears | 90046201001 |

**Example 5**. UMC created Umbrella records available in WHODrug.

### 6.1.4 NOS substances

NOS (Not Otherwise Specified) substances represent groups of substances and are generally created to be used as ingredients for records in WHODrug. All NOS records contain the abbreviation NOS in the end of the substance name.

| Drug name | Active substances |
|---|---|
| Dilyvita | Ascorbic acid, Retinol, Vitamin d nos |
| Staphymox Lb | Amoxicillin trihydrate, Cloxacillin sodium, Lactobacillus nos |

**Example 6**. NOS substances used as ingredients in WHODrug.

### 6.1.5 Non-unique names

In WHODrug identical drug names are available with different sets of ingredients, i.e. Non-unique names. A drug name can be Non-unique for several reasons, for example:

- The same drug name is used in different countries with different ingredients.
- The same drug name is used for different pharmaceutical forms which contain different sets of substances or variations of substances.
- A product has changed its composition without changing its trade name, leading to several records in WHODrug.

#### 6.1.5.1 Non-unique names in the B3 format

In the B3 format, Non-unique records are made unique by the addition of [INGREDIENT(S)] to the drug name. Therefore, every record in the B3 format is unique, even the Non-unique drug names. If the record has multiple ingredients, the ingredients will be ordered alphabetically, separated by a semicolon.

| Drug name | Active substance(s) |
|---|---|
| MENARIL [BETAHISTINE HYDROCHLORIDE] | Betahistine hydrochloride |
| MENARIL [KETOPROFEN] | Ketoprofen |

**Example 7**. A search for the drug name Menaril does not give an exact match in the B3 format of WHODrug. The reason is that this drug name is an example of a Non-unique drug name, hence bearing the appended [INGREDIENT(S)] in the end of the drug name.



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Note that generic Non-unique names do not have the appended [INGREDIENT(S)], to facilitate auto-encoding of these WHODrug records.

| Drug name | Active substance(s) |
|---|---|
| CALCIUM | Calcium |
| CALCIUM [CALCIUM GLUCONATE] | Calcium gluconate |
| CALCIUM [CALCIUM LACTATE GLUCONATE] | Calcium lactate gluconate |

**Example 8**. The generic Non-unique name Calcium does not have the appended [INGREDIENT(S)].

#### 6.1.5.2 Non unique names in the C3 format

In the C3 format, Non-unique drug names are not flagged in the same way as in the B3 format, as there is more information available for drug names in the C3 format, compared to the B3 format. The additional information levels available in the C3 format can therefore be used to differentiate between Non-unique names.

**Please note**

Non-unique drug names can be displayed differently than described above depending on your coding system.

## 6.2 Drug code

The WHODrug Drug code is an alphanumeric code, which is an aggregation of Drug Record Number (DrugRecNo), Sequence number 1 (Seq1) and Sequence number 2 (Seq2). The Drug code uniquely identifies a drug name and its substance(s) or an Umbrella record. The Drug code is used in both the B3 and the C3 formats and is the unique identifier in the B3 format.

### 6.2.1 Drug Record Number

The Drug Record Number (DrugRecNo) is the first part of the Drug code. It consists of six characters and uniquely identifies the substance, or combination of substances, regardless of variation or plant part.

WHODrug records sharing the same DrugRecNo contain the same substance(s) regardless of substance variation or product name.

| DrugRecNo | Seq1 | Seq2 | Active substance(s) | Active substance base(s) |
|---|---|---|---|---|
| 000200 | 01 | 001 | Paracetamol | Paracetamol |
| 132106 | 01 | 001 | Desogestrel;Estradiol | Desogestrel;estradiol |
| 013297 | 01 | 001 | Aesculus hippocastanum | Aesculus hippocastanum |
| 006460 | 01 | 001 | Insulin human | Insulin human |

**Example 9**. The DrugRecNo for Paracetamol, Desogestrel and estradiol, Aesculus hippocastanum, and Insulin human.



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### 6.2.2 Sequence number 1

Sequence number 1 (Seq1) is the second part of the Drug code, consisting of two characters. Seq1 is used to uniquely identify different substance variations.

In WHODrug, Seq1=01 always identifies a specific active substance base or a combination of substance bases. If Seq1 is higher than 01 it refers to variations of the substance(s).

WHODrug records sharing the same DrugRecNo and Seq1 contain the same variation of the same substance regardless of product name.

| DrugRecNo | Seq1 | Seq2 | Active substance(s) | Active substance base(s) |
|---|---|---|---|---|
| 000005 | 01 | 001 | Ampicillin | Ampicillin |
| 000005 | 02 | 001 | Ampicillin sodium | Ampicillin |
| 000005 | 03 | 001 | Ampicillin trihydrate | Ampicillin |
| 140048 | 01 | 001 | Metoprolol;Ramipril | Metoprolol;ramipril |
| 140048 | 02 | 001 | Metoprolol succinate;Ramipril | Metoprolol;ramipril |
| 140048 | 03 | 001 | Metoprolol tartrate;Ramipril | Metoprolol;ramipril |

**Example 10**. These first three records all share the same substance, Ampicillin, but in different base/salt variations; the base is represented by Seq1=01 while the sodium and trihydrate variations are represented by Seq1=02 and 03. All three records share the same DrugRecNo. The same logic also applies for the last three combination records.

For WHODrug records representing herbal medicinal products, Seq1 represents different plant parts or combination of plant parts.

| DrugRecNo | Seq1 | Seq2 | Active substance(s) | Active substance base(s) |
|---|---|---|---|---|
| 013297 | 01 | 001 | Aesculus hippocastanum | Aesculus hippocastanum |
| 013297 | 03 | 001 | Aesculus hippocastanum seed | Aesculus hippocastanum |
| 013297 | 11 | 001 | Aesculus hippocastanum bark | Aesculus hippocastanum |

**Example 11** The DrugRecNo and Seq1 for the herbal remedy Aesculus hippocastanum.

The value for Seq1 is generated for each specific DrugRecNo. Therefore, the same variation (e.g. sodium salts) can have different Seq1 values for different DrugRecNos.

### 6.2.3 Sequence number 2

Sequence number 2 (Seq2) is the third part of the Drug code, consisting of three characters. Seq2 uniquely identifies the name of the record in WHODrug. Records with Seq2=001 represent the substance or substance combination, and records with Seq2>001 represent product names and substance synonyms.



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| DrugRecNo | Seq1 | Seq2 | Product name | Active substance(s) | Active substance base(s) |
|---|---|---|---|---|---|
| 000005 | 01 | 001 | - | Ampicillin | Ampicillin |
| 000005 | 01 | 002 | Ampicin | Ampicillin | Ampicillin |
| 000005 | 02 | 001 | - | Ampicillin sodium | Ampicillin |
| 000005 | 02 | 003 | Polycillin-n | Ampicillin sodium | Ampicillin |
| 134314 | 01 | 001 | - | Ampicillin;Cloxacillin | Ampicillin;Cloxacillin |
| 134314 | 01 | 003 | Ampiclox | Ampicillin;Cloxacillin | Ampicillin;Cloxacillin |
| 134314 | 03 | 001 | - | Ampicillin trihydrate;Cloxacillin sodium | Ampicillin;Cloxacillin |
| 134314 | 03 | 010 | Zygoclox | Ampicillin trihydrate;Cloxacillin sodium | Ampicillin;Cloxacillin |

**Example 12**. The DrugRecNo, Seq1 and Seq2 for substances and product names containing Ampicillin, Ampicillin sodium, Ampicillin;Cloxacillin and Ampicillin trihydrate;Cloxacillin sodium.

## 6.3 C3 format specific concepts

### 6.3.1 Record Identifier (RID)

The Record ID (RID), formerly known as the Medicinal Product ID (MPID), is unique for each record in the C3 format and unlike the Drug code it is a serial number with no intrinsic meaning. Each RID uniquely identifies a specific combination of the following pieces of information:

- Drug name
- Name specifier
- Active substance
- Country
- Marketing authorisation holder
- Pharmaceutical form
- Strength

#### 6.3.1.1 Information levels

To allow for different degrees of specificity when coding using the C3 format, there are several information levels.



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| RID | Drug code | Drug name | Active substance(s) | Name specifier | Country | Marketing authorisation holder | Pharma-ceutical form | Strength |
|---|---|---|---|---|---|---|---|---|
| 67763 | 00544101003 | Bezalip | Bezafibrate | | *Unspecified* | *Unspecified* | *Unspecified* | *Unspecified* |
| 1264950 | 00544101003 | Bezalip | Bezafibrate | | Netherlands | *Unspecified* | *Unspecified* | *Unspecified* |
| 67762 | 00544101003 | Bezalip | Bezafibrate | | Netherlands | Roche | *Unspecified* | *Unspecified* |
| 6429827 | 00544101003 | Bezalip | Bezafibrate | | Netherlands | Roche | *Unspecified* | 400 mg |
| 69815 | 00544101003 | Bezalip | Bezafibrate | | Netherlands | Roche | Coated tablets | *Unspecified* |
| 69814 | 00544101003 | Bezalip | Bezafibrate | Retard | Netherlands | Roche | Coated tablets | 400 mg |

**Example 13.** The different RIDs identifies related records representing the same medicinal product but with different levels of specificity. The record with RID 67763 only holds information about the trade name and the active substance and can therefore be selected for coding when only the drug name is provided. This record also directly corresponds to the B3 format record. The RID 67762 on the other hand, has information about country and marketing authorisation holder and can therefore be selected when additional information about country and marketing authorisation holder is known.

**Please note**

Not all drug names in WHODrug have all information levels available.

### 6.3.2 Name specifier

Sometimes, one drug can have several variations of the same medicinal products with various trade name extensions, such as strength, pharmaceutical form or route of administration. For some legacy records, this extended information can be found in the Name specifier field. Seq2 (the last three characters in the Drug code) is set only by the drug name and not the Name specifier in WHODrug.

| Full name | Drug name | Name specifier |
|---|---|---|
| Sinemet 25/100 | Sinemet | 25/100 |
| Daktarin oral gel | Daktarin | oral gel |

**Example 14**. Examples of Name specifiers.

### 6.3.3 Archived record

The Archived record designation in WHODrug can reflect changes in medicinal product compositions, product withdrawal, or removal from market. Although UMC continually strives towards keeping WHODrug accurate and current, it should be noted that capturing the marketing status of medicinal products is not the main focus of WHODrug, and is done on requests from WHODrug users.

Sometimes pharma companies change the composition or MAH of a medicinal product without changing its trade name. In WHODrug this is reflected by the addition of a new record with the



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new composition of ingredients or MAH. Furthermore, the existing WHODrug record, representing the medicinal product with the old composition or MAH, will be flagged as Archived record. Both these records will remain in WHODrug since both compositions of the medicinal product still should be available for coding.

Archived record can be assigned to any level of RID specificity. If Archived record is assigned to the most specific RID (including trade name, substance, country of sale, MAH, form and strength), all linked less specific RIDs are also assigned with Archived record. An exception to this is when the less specific RIDs are also linked to other more specific RIDs. The least specific RID (including trade name and substance) is never assigned Archived record. In the C3 format, the Archived record designation is found in the additional feature file Archived Record.

| RID | Drug code | Drug name | Active substance | Country | MAH | Pharma-ceutical form | Strength | Archived record |
|---|---|---|---|---|---|---|---|---|
| 4287660 | 147779 02 002 | Troferit flow | Ambroxol hydrochloride; Dropropizine | | | | | |
| 4287659 | 147779 02 002 | Troferit flow | Ambroxol hydrochloride; Dropropizine | Mexico | | | | |
| 4969365 | 147779 02 002 | Troferit flow | Ambroxol hydrochloride; Dropropizine | Mexico | Carnot laboratorios, productos cientificos, s.a. de c.v. | | | Yes |
| 4969364 | 147779 02 002 | Troferit flow | Ambroxol hydrochloride; Dropropizine | Mexico | Carnot laboratorios, productos cientificos, s.a. de c.v. | Tablets | | Yes |
| 4287658 | 147779 02 002 | Troferit flow | Ambroxol hydrochloride; Dropropizine | Mexico | Chinoin, productos farmaceuticos | | | |
| 4287657 | 147779 02 002 | Troferit flow | Ambroxol hydrochloride; Dropropizine | Mexico | Chinoin, productos farmaceuticos | Liquids, syrups | | |
| 5507885 | 147779 02 002 | Troferit flow | Ambroxol hydrochloride; Dropropizine | Mexico | Chinoin, productos farmaceuticos | Tablets | | |
| 5507884 | 147779 02 002 | Troferit flow | Ambroxol hydrochloride; Dropropizine | Mexico | Chinoin, productos farmaceuticos | Tablets | 30 mg;30 mg | |

**Example 15.** Examples of Archived record designations for Troferit flow [Ambroxol hydrochloride;Dropropizine].



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# 7 The terminology of WHODrug Global Chinese

Since China became a member of ICH in 2017, the focus on drug regulatory capabilities has increased, emphasizing standardization within Chinese pharmaceutical and CRO industry. Aligned with the growing need to handle standardized drug data in the Chinese language, UMC provides the Chinese version of WHODrug Global to enable accurate and efficient drug coding without the need for manual and time-consuming translations into English.

## 7.1 Drug information

In WHODrug Global Chinese drug names are shown in Chinese for drugs approved in China. In addition, active substance(s), ATC code text, country, marketing authorisation holder and pharmaceutical form are also displayed in Chinese. All Chinese records have an equivalent English drug name in WHODrug Global connected by the Drug code.

**B3**

| Drug code | Drug name | Active substance(s) | ATC |
|---|---|---|---|
| 00002701A7V | 阿容 | 乙酰水杨酸 | A01AD, 其它口腔局部用药物 *official*<br>B01AC, 血小板凝固抑制剂, 不包括肝素类 *official*<br>M02AC, 水杨酸衍生物类药物 *umc-assigned*<br>N02BA, 水杨酸和其衍生物 *official* |

**C3**

| Drug code | Drug name | Active substance(s) | ATC | Country of sales | Marketing authorisation holder | Pharmaceutical form | Strength | RID |
|---|---|---|---|---|---|---|---|---|
| 00002701A7V | 阿容 | 乙酰水杨酸 | B01AC, 血小板凝固抑制剂, 不包括肝素类 *official* | 中国 | 山东鲁抗医药集团鲁亚有限公司 | 包衣片,肠溶 | 0.1g | 4726294 |

**Example 16.** The record 阿容 in the B3 format and in the C3 format in WHODrug Global Chinese.

For drugs approved outside of China, the active substance(s), ATC code text, country and pharmaceutical form are displayed in Chinese, while the trade name and marketing authorisation holder are displayed in English with Latin characters.

**B3**

| Drug code | Drug name | Active substance(s) | ATC |
|---|---|---|---|
| 00002701121 | Trombyl | 乙酰水杨酸 | A01AD, 其它口腔局部用药物 *official*<br>B01AC, 血小板凝固抑制剂, 不包括肝素类 *official*<br>M02AC, 水杨酸衍生物类药物*umc-assigned*<br>N02BA, 水杨酸和其衍生物 *official* |



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**C3**

| Drug code | Drug name | Active substance(s) | ATC | Country | Marketing authorisation holder | Pharmaceutical form | Strength | RID |
|---|---|---|---|---|---|---|---|---|
| 00002701121 | Trombyl | 乙酰水杨酸 | B01AC, 血小板凝固抑制剂, 不包括肝素类 *official* | 瑞典 | Pfizer | 片剂 | 75 mg | 2881994 |

**Example 17.** The record Trombyl in the B3 format and in the C3 format in WHODrug Global Chinese.

## 7.2 Pinyin names

Chinese products can be included in WHODrug with the phonetic spelling of the drug name using Latin characters based on the official romanisation system for Standard Chinese, Pinyin.

When a Chinese drug name is lacking a corresponding official English name, it is included as a translated Pinyin name in WHODrug Global. The Pinyin name is connected to the Chinese drug name by the Drug code. For example, the drug name 可迪 does not have a corresponding official English name, so it is included in Pinyin in WHODrug Global as 可迪(Ke di). The Pinyin name is displayed with a space between every character, e.g. 可(Ke)迪(Di) is displayed as Ke Di in WHODrug. This spacing rule only applies for Chinese products approved in China. Products outside China with a Pinyin name is displayed as the name on the package, with no additional spaces.

A list of common pharmaceutical forms translated from Chinese to Pinyin and English can be found in Appendix 1.

## 7.3 Non-unique names

For some Chinese products the Pinyin name is the same although the Chinese characters are different. Therefore, these records have non-unique names in WHODrug Global but are unique in WHODrug Global Chinese. Non-unique product records in the B3 format are made unique by the addition of [INGREDIENT(S)] to the drug name.



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**WHODrug Global**

| Drug code | Drug name | Active substance(s) | ATC |
|---|---|---|---|
| 00048102382 | KE DI [DEXTROMETHORPHAN HYDROBROMIDE] | Dextromethorphan hydrobromide | R05DA, Opium alkaloids and derivatives *umc-assigned* |
| 00876101024 | KE DI [NISOLDIPINE] | Nisoldipine | C08CA, Dihydropyridine derivatives *official* |

**WHODrug Global Chinese**

| Drug code | Drug name | Active substance(s) | ATC |
|---|---|---|---|
| 00048102382 | 可迪 | 氢溴酸美沙芬 | R05DA, 鸦片生物碱及其衍生物类 *umc-assigned* |
| 00876101024 | 可谛 | 尼索地平 | C08CA, 二氢吡啶衍生物 *official* |

**Example 18**. Non-unique trade names become unique.

## 7.4 Umbrella records

All Umbrella records are displayed in Chinese. The Drug code is the same in WHODrug Global Chinese as in WHODrug Global.



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# 8 ISO IDMP data and mappings to WHODrug

IDMP (Identification of Medicinal Products) is a set of five standards developed by the International Organization for Standardization (ISO) to create a universal framework of structured, coded data that uniquely identify and describe all key aspects of medicinal products. This allows for harmonisation of medicinal product information at a global level and provides the guidance to creating global identifiers that can be used to describe pharmaceutical products. The unique global Pharmaceutical Product Identifiers (PhPIDs) identify the pharmaceutical product which is the abstract description of a medicinal product, which includes the substance, strength and dose.

With ISO IDMP standards, medicinal products will be treated consistently throughout the entire medicinal life cycle, from development to post-marketing use. At the research and development stages of the medicinal product lifecycle, information can be streamlined by following the ISO IDMP standards during clinical trials. As marketing authorisation is approved, using the ISO IDMP standards make post-marketing processes more harmonised to help many stakeholders when medicines are used, in monitoring safety data, when taking regulatory actions and performing new medicinal developments.

To maintain the benefits of using WHODrug Global, with its unique structure and linkage to established classifications, while facilitating the use of ISO IDMP data, WHODrug RIDs are mapped to global PhPIDs.

## 8.1 Pharmaceutical Product ID (PhPID)

The Pharmaceutical Product ID (PhPID) is generated according to ISO 11616, based on the Global Substance Identifier (GSID) which identifies the substance, and dose form and strength. PhPIDs are available as “sets” of four, generated for different levels of granularity with level 1 representing the substance, level 2 representing the substance and strength, level 3 representing substance and form, and level 4 representing substance, form and strength. PhPIDs are alpha-numerical and consist of 21 characters, starting with a seven-character qualifier, UTPHPID.

| B3 record | C3 RID | Country of Sale | MAH | Form | Strength | PhPID |
|---|---|---|---|---|---|---|
| | 56003 | - | - | - | - | UTPHPID475L8E4RDB2F4Y (PhPID lvl 1) |
| **Humira** | 1344826 | Brazil | - | - | - | UTPHPID475L8E4RDB2F4Y (PhPID lv 1) |
| **Drug code: 01612901002** | 3050353 | Brazil | Abbvie | - | - | UTPHPID475L8E4RDB2F4Y (PhPID lvl 1) |
| | 6661213 | Brazil | Abbvie | - | 40 mg | UTPHPID5T0ZSM4JBZ5ZY2 (PhPID lvl 2) |
| **Ingredient: Adalimumab** | 3258216 | Brazil | Abbvie | VIALS, SUBCUTANEOUS | - | UTPHPID462S58KER7GD4B (PhPID lvl 3) |
| **ATC: L04AB** | 3050351 | Brazil | Abbvie | VIALS, SUBCUTANEOUS | 40 mg | UTPHPID1P9RJ4XH086JEF (PhPID lvl 4) |

**Example 19.** The WHODrug C3 record Humira mapped to corresponding PhPID levels.

In WHODrug, PhPIDs are mapped to WHODrug RIDs with matching levels of product detail. This means that multiple WHODrug RIDs can share the same PhPID. In the example of Humira,



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the first 3 RIDs with different specificity of medicinal product information (56003, 1344826, 3050353) all are mapped to the same PhPID level 1 UTPHPID475L8E4RDB2F4Y. Other RIDs identifying the same pharmaceutical product (adalimumab in same strength and form) share the same PhPID, even if the product name, country of sales and MAH may differ.

Because the ISO IDMP standard is more granular in definition than WHODrug, there may also be instances where a single WHODrug RID may be mapped to more than one PhPID.

## 8.2 Global Substance Identifiers (GSIDs)

GSID is a unique identifier of substances used in medicinal products and is assigned according to ISO 11238. Pharmaceutical products containing several active ingredients will use the individual GSID for each active ingredient to generate a PhPID. GSIDs are alpha-numerical and consist of 19 characters, starting with a six character qualifier UTGSID.

## 8.3 Strength

The strength used in generating the PhPID captures the quantitative expression of a product's strength using a combination of numerical value and unit of measurement, standardized according to Unified Code for Units of Measure (UCUM) in alignment with ISO 11240.

## 8.4 Dose Form

In generating the global PhPID, common dose form attributes are used to express dose form to enable mapping between regional dose form terminologies, instead of a singular dose form terminology. Four dose form attributes are used to describe a pharmaceutical product's form and administration according to ISO 11239.

### 8.4.1 Basic dose form

Generalised version of the pharmaceutical dose form, used to group together related pharmaceutical dose forms (e.g. tablet, capsule, powder).

### 8.4.2 Administration method

General method by which a pharmaceutical product is intended to be administered to the patient (e.g. application, inhalation, injection). A PhPID can have up to 3 administration methods, depending on the pharmaceutical product.

### 8.4.3 Intended site

General body site at which a pharmaceutical product is intended to be administered (e.g. auricular, ocular, oral). A PhPID can have up to 3 intended sites, depending on the pharmaceutical product.

### 8.4.4 Release characteristics

Description of the timing by which an active ingredient is made available in the body after administration of the pharmaceutical product, in comparison with a conventional, direct release of the active ingredient (e.g. delayed, prolonged, conventional, modified).



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# 9 Accessing WHODrug data

WHODrug data can be accessed by downloading the WHODrug files or through our online search application Insight.

## 9.1 WHODrug Insight

WHODrug Insight is the online search application that provides instant access to the data available in WHODrug in both B3 and C3 terminology formats. It is possible to view WHODrug data in various versions, which provides a better understanding of the differences between terminology versions. Insight also incorporates WHODrug Upcoming data, which reflects how the terminology would look like if released today. The upcoming data is updated on a daily basis and may be subject to change at any time until the data lock point prior to the next release.

The search functionality in Insight supports several search criteria, such as drug name, ingredient, ATC code, as well as combinations of search criteria, to enable efficient coding of drug names and drug classes. The search result can also be exported to an Excel file.

The WHODrug Standardised Drug Groupings (SDGs) are also available in Insight. This feature enables the identification of drug lists based on various properties of interest and facilitates the creation of organisation specific medication lists of interest, called Customized Drug Groupings (CDGs). Both the SDGs and the CDGs can be exported to an Excel file.

Mappings between WHODrug and other data, such as ISO IDMP data, as well as mappings to local datasets like CRT Japan and Link Korea, can also be accessed via Insight.

## 9.2 WHODrug Global file packages

WHODrug is a relational database, provided in data files independent of any specific database system. In order to use the information, the files need to be implemented in a coding application, which could be an in-house system, various Electronic Data Capture (EDC) systems, Adverse Event database systems and Concomitant Medication Coding tools that are available on the market. To facilitate the setup and simplify the use of the terminology, it is provided in a file package and delivered to the customer download area of the appointed delivery contact within your organisation. WHODrug Global and WHODrug Global Chinese are provided as two separate but complete terminology packages. Both packages employ the same file structure.

The WHODrug Global files and WHODrug Global Chinese files are available in CSV format, to facilitate the loading of information depending on system setup.

The WHODrug terminology files are arranged in two package types: standard files and additional features files. The terminology content is stored in the standard files, whereas additional features files are developed to accommodate the need for flexibility and thus allow for a customised setup of WHODrug. For a more technical description, please refer to the file descriptions included in each package.

### 9.2.1 WHODrug Global standard files

The standard files of the terminology contain basic information about the drug names. In the core files, you can find information such as the Drug code, the drug name, active substance(s), information about the type of record and number of ingredients. The ATC classification, marketing authorisation holder, pharmaceutical form and strength are also examples of information found in the standard files.



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### 9.2.2 WHODrug Global Chinese standard files

The Chinese standard files of the terminology contain the same basic information about the drug names as in WHODrug Global, but the drug name, active substance(s), country, marketing authorisation holder, pharmaceutical form and ATC text are displayed in Chinese.

A format description and test cases are provided together with the files to ensure correct implementation.

### 9.2.3 ISO IDMP data files and mapping to WHODrug

The main files include PhPID and substance information, mapping to WHODrug RIDs and additional code table files label codes with names.

The ISO IDMP data is a set of main files that describe PhPIDs, their hierarchical levels, product attributes, and substance compositions, together with their corresponding mappings to the WHODrug identifiers. It includes files linking each PhPID to WHODrug RIDs and UMCPIDs, defining PhPID levels, and detailing dose form attributes. Additional files describe the substances and strengths associated with each PhPID through GSIDs, and provide the corresponding substance names.

### 9.2.4 WHODrug Global additional features files

The additional feature files have been developed over time as the need for flexibility has grown. Below is a brief description of the additional features available.

#### 9.2.4.1 Cumulative changes

Cumulative changes is available for both B3 and C3 formats. The file includes information about deleted and replaced Drug codes in the B3 format, and deleted and replaced RIDs in the C3 format. The files also include which terminology version the Drug code and RID was last released.

#### 9.2.4.2 DDA Exclusive

This file makes it possible to map the Drug code to the ATC assignments with higher precision when using the B3 format. This file is further described in the chapter “ATC assignments in the B3 format”.

#### 9.2.4.3 BNA Translation

BNA Translation is included in the additional feature file package for the B3 format. BNA translation includes information about Spanish or Portuguese language code and corresponding substance name. Chemical Abstracts Service (CAS) number is used as unique substance identifier and connects to WHODrug ING (ingredient) and DD source file.

#### 9.2.4.4 SUN Translation

SUN Translation is included in the additional feature file package for the C3 format. SUN translation includes information about Spanish or Portuguese language code and corresponding substance name. CAS number is used as a unique substance identifier and connects to the WHODrug ING (ingredient) and DD source file. Substance_Id is used as a unique substance identifier and connects to the WHODrug ING (ingredient) file.

#### 9.2.4.5 UNII drug list

Contains FDA Unique Ingredient Identifier (UNII) codes and their corresponding Drug codes in WHODrug.

#### 9.2.4.6 Archived Record

Archived Record is included in the additional feature file package for the C3 forrmat. Archive Record is a table of all the RIDs that have been marked Archived Record.



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### 9.2.5 WHODrug Global Chinese additional feature files

One additional feature file is available for WHODrug Global Chinese in addition to the ones included in WHODrug Global.

#### 9.2.5.1 English Drug Name

This file includes the active substance names in English, and Chinese product names spelled with Pinyin that are included in WHODrug Global. This file is intended to facilitate data submission and analysis in different languages.

### 9.2.6 WHODrug Standardised Drug Groupings files

The Standardised Drug Groupings (SDG) files consists of four main files. The files include information about the SDG groups, the group content, the scope and ATC codes.

#### 9.2.6.1 SDG

This file includes information about the parent and subgroups of each SDG. The file also sets out the WHODrug version in which the SDG was first released, and when it was last changed.

#### 9.2.6.2 SDGContent

Information about Drug codes that are included in the SDG can be found in this file. The drug name, number of ingredients, and scope is included for all Drug codes.

#### 9.2.6.3 SDGScope Lx

Narrow or broad scope is specified in this file.

#### 9.2.6.4 SDGATCPreferred

This file contains all WHODrug Drug codes that are included in any SDG, and their assigned ATC code(s).



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# 10 ATC classification

The Anatomical Therapeutic Chemical (ATC) classification is a hierarchical classification system which classifies active substances according to the organ or system on which they act, as well as their therapeutic, pharmacological and chemical properties. The ATC classification is integrated in WHODrug.

## 10.1 WHO Collaborating Centre for Drug Statistics Methodology ATC classification system

The ATC classification system is maintained by the WHO Collaborating Centre for Drug Statistics Methodology, in Oslo, Norway (WHO CC, Oslo). The WHO International Working Group for Drug Statistics Methodology includes new active substances in the ATC classification hierarchy on request from users of the system. Users might include marketing authorisation holders, regulatory agencies, and researchers.

The ATC hierarchy consists of five levels. The first level in the ATC system is represented by 14 anatomical/pharmacological main groups. The second further classifies active substances according to pharmacological/therapeutic properties. The third and the fourth ATC levels are chemical/pharmacological/therapeutic subgroups, and the fifth represents active substances.

| ATC code | ATC Text |
|---|---|
| N | Nervous system (First level, anatomical main group) |
| N02 | Analgesics (Second level, therapeutic subgroup) |
| N02A | Opioids (Third level, pharmacological subgroup) |
| N02AA | Natural opium alkaloids (Fourth level, chemical subgroup) |
| N02AA01 | Morphine (Fifth level, chemical substance) |

**Example 20**. The full ATC classification of Morphine.

The WHO Collaborating Centre for Drug Statistics Methodology mainly use the ATC system for drug utilisation research. It is not primarily intended for identifying all the indications a specific medicinal product might have. Furthermore, a medicinal product is generally only assigned more than one ATC code if it is available in two or more non-overlapping strengths, if it is available in different pharmaceutical forms for topical and systemic use, or if it is available in two or more routes of administration with clearly different therapeutic uses.

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For more detailed information and to learn more about the ATC classification system visit http://www.whocc.no/.



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Box 1051, SE-751 40 Uppsala, Sweden
Email: support@who-umc.org, www.who-umc.org

## 10.2 Herbal ATC

The Herbal ATC (HATC) system was originally developed by UMC and provides a scientific framework for a harmonised, global nomenclature and therapeutic classification of herbal remedies and combinations thereof. It follows the same principles as the conventional ATC classification system; the ATC and the HATC systems are closely intertwined. HATC codes can be regarded as additional branches in the ATC hierarchy. Analysis on HATC codes is therefore possible using the same methodology as for the ATC system.

The first level in the HATC system is the same as in the ATC system, the 14 anatomical/pharmacological main groups. The following levels also follow the hierarchy and structure for the ATC system with the addition of herbal specific codes, to better identify herbal specific properties.

| HATC code | HATC text |
|---|---|
| A | Alimentary tract and metabolism (First level, anatomical main group) |
| A02 | Drugs for acid related disorders (Second level, therapeutic subgroup) |
| A02W | Herbal remedies for treatment of peptic ulcer (Third level, pharmacological subgroup) |
| A02WA | Herbal remedies for treatment of peptic ulcer containing saponins (Fourth level, chemical subgroup) |

**Example 21**. The hierarchy of A02WA in the HATC system.

## 10.3 ATC assignments in WHODrug

The purpose of ATC assignments in WHODrug is to classify medications according to their intended use. The ATC classification is primarily based on the guidelines for ATC classification, as provided by the WHO CC, Oslo. All records in WHODrug are assigned at least one ATC code, preferably a fourth level ATC code. If, for any reason, it is not possible to assign a WHODrug record a fourth level ATC code, less specific (i.e. first, second or third level) ATC codes are assigned.

ATC level five assignments are not included in WHODrug, for one main reason: ATC level five is used to identify active substances. Since not all active substances have an officially assigned ATC code, WHODrug utilises another feature to uniquely identify active substances – the Drug code. Marketing authorisation holders for medicinal products registered in the EU are currently obliged to submit the most specific ATC code (i.e. the fifth level) in xEVMPD submission according to Article 57. To facilitate the submission of ATC level five information WHODrug Cross Reference ATC 5 has been developed. WHODrug Cross Reference ATC 5 consists of a mapping table which translates the Drug code in WHODrug to ATC level five ATC assignments.

**Please note**

WHODrug Cross Reference ATC 5 does not contain HATC level five codes.



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In WHODrug, conventional and herbal remedies are seamlessly integrated into one collection of drug names. Furthermore, each record is assigned at least one conventional or HATC code, depending on the origin of the active substance(s) (i.e. conventional or herbal).

### 10.3.1 Official and UMC assigned ATC codes

The ATC classification of records in WHODrug is primarily based on the guidelines for ATC classification, as provided by the WHO CC, Oslo. The ATC codes assigned to an active substance in the ATC index is always assigned the same ATC code in WHODrug. These ATC classifications as in the official ATC index are indicated with an official flag.

| Drug name | Active substance(s) | ATC |
|---|---|---|
| Pamol | Paracetamol | N02BE, Anilides *official* |

**Example 22**. Paracetamol is assigned the official ATC code N02BE in WHODrug.

In addition to official ATC assignments, there are also UMC-assigned ATC codes available in WHODrug. UMC-assigned ATC codes are included due to the specific use cases of WHODrug, when official ATC codes are not sufficient for efficient analysis of medicinal products and their properties. UMC-assigned ATC codes are the same codes as found in the official ATC classification system and assigned to active substances without an official ATC code in the ATC Index, and/or indicate alternative use or properties of a specific medicinal product. The ATC codes in WHODrug primarily reflect labelled indication(s), which means that additional ATC codes for off-label indications are generally not available.

| Drug name | Active substance(s) | ATC |
|---|---|---|
| Sildenafil | Sildenafil | C02KX, Antihypertensives for pulmonary arterial hypertension *umc-assigned*<br>G04BE, Drugs used in erectile dysfunction *official* |

**Example 23**. Sildenafil has only one official ATC assignment, G04BE, which is also flagged as official in WHODrug. In addition, to highlight alternative indications, this active substance is also assigned with C02KX, which is flagged as UMC-assigned.

Combinations of active substances are rarely assigned official ATC codes. Therefore, most multi-ingredient records in WHODrug have UMC-assigned ATC codes.

### 10.3.2 UMC created ATC codes

In addition to the codes in the ATC and HATC hierarchy, there are a few UMC-created ATC codes added to enable specific WHODrug use. As these ATC codes are added by UMC they are all flagged as UMC-assigned. Currently, there are three UMC-created ATC codes:

#### 10.3.2.1 V90: Unspecified herbal and traditional medicine

In addition to HATC code assignments, all herbal remedies are also assigned the UMC-developed ATC code V90: Unspecified herbal and traditional medicine. V90 can be used to identify herbal and traditional medicines (e.g. TCM, Kampo) in WHODrug.

#### 10.3.2.2 V91: Homeopathic preparation

There are a few homeopathic remedies in WHODrug. These remedies are assigned the UMC-developed ATC code V91, and no additional ATC codes. For homeopathic products containing only herbal ingredients, V90 is also assigned.



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#### 10.3.2.3 V98: Investigational drug

The UMC-created ATC code V98 was introduced to facilitate the identification of non-approved active substances currently in clinical trials. The ATC assignment V98 does not exclude additional ATC assignments. If an active substance assigned V98 receives marketing approval and/or is included as an ingredient of a WHODrug record (i.e. a trade name), the ATC V98 will be removed.

**Please note**

The codes V98 and V91 are assigned to new investigational drugs and homeopathic remedies as they are included in WHODrug. Historical records may not have this assigned ATC code, even though they fulfil the inclusion criteria. If you think that such an ATC assignment for a specific WHODrug record is missing, please make use of the Change Request application.

### 10.3.3 Inherited ATC assignments in WHODrug

Records connected by the Drug code in WHODrug are also connected by ATC assignments. The substance base always inherits all the ATC codes that are assigned to any substance variation(s) sharing the same DrugRecNo. For single substances the active substance variation(s) does not inherit the ATC codes of the substance base. However, for substance combinations, the substance base and all its substance variations share the same ATC codes.

| Drug code | Active substance(s) | B3 format ATC code(s) |
|---|---|---|
| 001092 01 001 | Ibuprofen | C01EB, Other cardiac preparations *official*<br>D10AX, Other anti-acne preparations for topical use *umc-assigned*<br>D11AH, Agents for dermatitis, excluding corticosteroids *umc-assigned*<br>G02CC, Antiinflammatory products for vaginal administration *official*<br>M01AE, Propionic acid derivatives *official*<br>M02AA, Antiinflammatory preparations, non-steroids for topical use *official*<br>R02AX, Other throat preparations *official* |
| 001092 02 001 | Ibuprofen lysinate | C01EB, Other cardiac preparations *official*<br>G02CC, Antiinflammatory products for vaginal administration *official*<br>M01AE, Propionic acid derivatives *umc-assigned*<br>M02AA, Antiinflammatory preparations, non-steroids for topical use *official* |
| 001092 04 001 | Ibuprofen piconol | C01EB, Other cardiac preparations *official*<br>D10AX, Other anti-acne preparations for topical use *umc-assigned*<br>D11AH, Agents for dermatitis, excluding corticosteroids *umc-assigned*<br>G02CC, Antiinflammatory products for vaginal administration *official*<br>M01AE, Propionic acid derivatives *umc-assigned*<br>M02AA, Antiinflammatory preparations, non-steroids for topical use *official* |
| 001092 05 001 | Ibuprofen sodium | C01EB, Other cardiac preparations *official*<br>M01AE, Propionic acid derivatives *umc-assigned*<br>M02AA, Antiinflammatory preparations, non-steroids for topical use *official* |
| 001092 07 001 | Ibuprofen isobutanolammonium | G02CC, Antiinflammatory products for vaginal administration *umc-assigned* |

**Example 24**. The substance base Ibuprofen inherits all the ATC assignment from all its substance variations.



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| Drug code | Active substance(s) | B3 format ATC code(s) |
|---|---|---|
| 131417 01 001 | Chloramphenicol;Hydrocortisone | D07CA, Corticosteroids, weak, combinations with antibiotics *umc-assigned*<br>S01CA, Corticosteroids and antiinfectives in combination *umc-assigned* |
| 131417 02 001 | Chloramphenicol palmitate;Hydrocortisone | D07CA, Corticosteroids, weak, combinations with antibiotics *umc-assigned*<br>S01CA, Corticosteroids and antiinfectives in combination *umc-assigned* |
| 131417 03 001 | Chloramphenicol;Hydrocortisone caproate | D07CA, Corticosteroids, weak, combinations with antibiotics *umc-assigned*<br>S01CA, Corticosteroids and antiinfectives in combination *umc-assigned* |
| 131417 04 001 | Chloramphenicol;Hydrocortisone caprylate | D07CA, Corticosteroids, weak, combinations with antibiotics *umc-assigned*<br>S01CA, Corticosteroids and antiinfectives in combination *umc-assigned* |

**Example 25**. The substance base combination Chloramphenicol;Hydrocortisone and all its substance variations all share the same ATC codes.

### 10.3.4 ATC assignments in the B3 format

In the B3 format the ATC assignments of the substance names are always given to all drug names sharing the same DrugRecNo and Seq1. In other words; all trade names are assigned the same ATC codes as the active substance or substance combination, regardless of pharmaceutical form, strength etc.

| Drug code | Drug name | Active substance(s) | B3 format ATC code(s) |
|---|---|---|---|
| 000027 01 001 | Acetylsalicylic acid | Acetylsalicylic acid | A01AD, Other agents for local oral treatment *official*<br>B01AC, Platelet aggregation inhibitors excl. heparin *official*<br>M02AC, Preparations with salicylic acid derivatives *umc-assigned*<br>N02BA, Salicylic acid and derivatives *official* |
| 000027 01 004 | Aspirin | Acetylsalicylic acid | A01AD, Other agents for local oral treatment *official*<br>B01AC, Platelet aggregation inhibitors excl. heparin *official*<br>M02AC, Preparations with salicylic acid derivatives *umc-assigned*<br>N02BA, Salicylic acid and derivatives *official* |
| 000027 01 121 | Trombyl | Acetylsalicylic acid | A01AD, Other agents for local oral treatment *official*<br>B01AC, Platelet aggregation inhibitors excl. heparin *official*<br>M02AC, Preparations with salicylic acid derivatives *umc-assigned*<br>N02BA, Salicylic acid and derivatives *official* |
| 131935 01 001 | Acetylsalicylic acid;Glycine | Acetylsalicylic acid;Glycine | B01AC, Platelet aggregation inhibitors excl. heparin *umc-assigned*<br>N02BA, Salicylic acid and derivatives *umc-assigned* |
| 131935 01 025 | Pluscard | Acetylsalicylic acid;Glycine | B01AC, Platelet aggregation inhibitors excl. heparin *umc-assigned*<br>N02BA, Salicylic acid and derivatives *umc-assigned* |
| 131935 01 026 | Alka Prim | Acetylsalicylic acid;Glycine | B01AC, Platelet aggregation inhibitors excl. heparin *umc-assigned*<br>N02BA, Salicylic acid and derivatives *umc-assigned* |

**Example 26**. ATC assignment for various Acetylsalicylic acid and Acetylsalicylic acid;Glycine records in WHODrug, as displayed in the B3 format.

The additional feature file DDA Exclusive makes it possible to retrieve ATC codes for each specific trade name in WHODrug, even when using the B3 format.



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Box 1051, SE-751 40 Uppsala, Sweden
Email: support@who-umc.org, www.who-umc.org

### 10.3.5 ATC assignments in the C3 format

The C3 format allows for a higher precision of ATC assignments than in the B3 format since the specific ATC assignments for each drug name are displayed in the C3 format.

Single- and multi-ingredient preparations can be assigned different ATC codes depending on the pharmaceutical form, strength and indication. In other words, specific ATCs are assigned for the different trade names containing the same active substance(s) while the substance base name is always assigned all ATC codes for that specific DrugRecNo.

| Drug code | Drug name | Active substance(s) | C3 format ATC code(s) |
|---|---|---|---|
| 000027 01 001 | Acetylsalicylic acid | Acetylsalicylic acid | A01AD, Other agents for local oral treatment *official*<br>B01AC, Platelet aggregation inhibitors excl. heparin *official*<br>M02AC, Preparations with salicylic acid derivatives *umc-assigned*<br>N02BA, Salicylic acid and derivatives *official* |
| 000027 01 004 | Aspirin | Acetylsalicylic acid | N02BA, Salicylic acid and derivatives *official* |
| 000027 01 121 | Trombyl | Acetylsalicylic acid | B01AC, Platelet aggregation inhibitors excl. heparin *official* |
| 131935 01 001 | Acetylsalicylic acid;Glycine | Acetylsalicylic acid;Glycine | B01AC, Platelet aggregation inhibitors excl. heparin *umc-assigned*<br>N02BA, Salicylic acid and derivatives *umc-assigned* |
| 131935 01 025 | Pluscard | Acetylsalicylic acid;Glycine | B01AC, Platelet aggregation inhibitors excl. heparin *umc-assigned* |
| 131935 01 026 | Alka Prim | Acetylsalicylic acid;Glycine | N02BA, Salicylic acid and derivatives *umc-assigned* |

**Example 27**. ATC assignment for various Acetylsalicylic acid and Acetylsalicylic acid;Glycine records in WHODrug, as displayed in the C3 format.



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Email: support@who-umc.org, www.who-umc.org

# 11 Changes to the WHODrug data

## 11.1 WHODrug Change Analysis Tool

UMC continually works to keep the content of WHODrug as comprehensive, up-to-date and correct as possible. Therefore, changes and updates to the WHODrug data are a feature of the terminology. UMC recommends the use of WHODrug Change Analysis Tool (CAT) to track these changes and their possible impact.

WHODrug CAT enables users to analyse the impact of data changes from one version of WHODrug to another. It is possible to compare any two versions of WHODrug (from 2006 and onwards) without having to go via in-between versions in the process. It is also possible to upload specific data (e.g. synonym lists and study data) and retrieve a full WHODrug B3 format change analysis including a list of new exact matches for any uploaded verbatim terms. WHODrug CAT can also be used for analysis of changes in the Standardised Drug Groupings (SDGs). CAT can be accessed via the UMC website.

## 11.2 WHODrug Change Request

If a medicinal product does not appear in WHODrug, a change request can be submitted via the WHODrug Change Request application, accessed at the UMC website. It is also possible to suggest a modification to an existing record, ATC changes and new SDGs in the application. Basic information such as trade name, active substance(s) and a reference should be included in the request. If the request is accepted, information about the drug is provided via a link to WHODrug Upcoming data in Insight. If for any reason it is not possible for UMC to include the requested record in WHODrug, a placeholder record is often suggested. Submitting change requests is free of charge for all WHODrug users. A first response from UMC is usually received within three business days, however, complex requests and large batch requests may require longer time to process.

If you have questions regarding an existing record in WHODrug, please include identifying information about the record (e.g. Drug code) in the request and e-mail us at: support@who-umc.org



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Box 1051, SE-751 40 Uppsala, Sweden
Email: support@who-umc.org, www.who-umc.org

Uppsala Monitoring Centre (UMC) is an independent, self-funded, non-profit foundation established in 1978 dedicated to safer use of medicines and vaccines. Through an agreement between the Government of Sweden and the World Health Organization (WHO), UMC operates the WHO Programme for International Drug Monitoring, supporting over 180 member countries and regions in strengthening safety surveillance, and maintains VigiBase, the WHO global database of adverse event reports. In addition, UMC provides international standards and related digital solutions for secure exchange of pharmacovigilance data, including a global medicine and vaccine terminology for identification of medicinal products. With around 200 staff, UMC advances the science of pharmacovigilance and transforms its practice through technological innovation. www.who-umc.org

**Uppsala Monitoring Centre (UMC)**
Box 1051, SE-751 40 Uppsala, Sweden
Tel: +46 18-65 60 60 **www.who-umc.org**



**Uppsala Monitoring Centre (UMC)**
Box 1051, SE-751 40 Uppsala, Sweden
Email: support@who-umc.org, www.who-umc.org

# Appendix 1

## Chinese form list

Chinese product names may include a pharmaceutical form in pinyin. However, in WHODrug, the form is always removed from the product name. For products marketed in China, the form is added in the Name specifier field. To help WHODrug users understand the meaning of common pinyin forms in Chinese products, we have compiled a list of Chinese forms included in the Name specifier field in WHODrug. The meaning of a Chinese form name can vary depending on its actual usage in the product. Please note that this list is for reference only and may not cover all types of Chinese forms or may have more meanings than listed.

| Form in Chinese | Form in Pinyin | Equals to English |
|---|---|---|
| 喷雾剂 | Pen wu ji | Spray |
| 鼻喷雾剂 | Bi pen wu ji | Nasal Spray |
| 鼻喷雾剂 | Bi yong pen wu ji | Nasal Spray |
| 胶 | Jiao | Glue |
| 胶囊 | Jiao nang | Capsules |
| 软胶囊 | Ruan jiao nang | Soft capsules |
| 肠溶胶囊 | Chang rong jiao nang | Enteric-coated capsules |
| 缓释胶囊 | Huan shi jiao nang | Sustained release capsules |
| 滴剂 | Di ji | Drops |
| 滴丸 | Di wan | Dripping pills |
| 滴眼液 | Di yan ye | Eye drops |
| 滴鼻液 | Di bi ye | Nasal drops |
| 散 | San | Powder |
| 粉 | Fen | Powder |
| 粉针剂 | fen zhen ji | Powder injection |
| 冻干粉 | Dong gan fen | Freeze-dried powder |
| 混悬剂 | Hun xuan ji | Suspension |
| 干混悬剂 | Gan hun xuan ji | For suspension |
| 含化滴丸 | Han hua di wan | Orally disintegrating dripping pills |
| 冲剂 | Chong ji | Granules |
| 颗粒 | Ke li | Granules |
| 合剂 | He ji | Liquid mixture |
| 酒 | Jiu | Alcohol |
| 药酒 | Yao jiu | Medical alcohol |
| 糖浆 | Tang jiang | Syrup |



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| | | |
|---|---|---|
| 口服液 | Kou fu ye | Oral solution |
| 口服溶液 | Kou fu rong ye | Oral solution |
| 口服乳液 | Kou fu ru ye | Oral lotion |
| 粘合剂 | Nian he ji | Sealant kit |
| 凝胶 | Ning jiao | Gel |
| 片 | Pian | Tablets |
| 含片 | Han pian | Buccal tablets |
| 泡腾片 | Pao teng pian | Effervescent tablet |
| 缓释片 | Huan shi pian | Sustained release tablets |
| 咀嚼片 | Ju jue pian | Chewable tablet |
| 分散片 | Fen san pian | Dispersible tablets |
| 控释片 | Kong shi pian | Controlled-release tablets |
| 肠溶片 | Chang rong pian | Enteric-coated tablets |
| 肠溶缓释片 | Chang rong huan shi pian | Enteric-coated sustained-release tablets |
| 气雾剂 | Qi wu ji | Aerosol |
| 溶液 | Rong ye | Solution |
| 栓 | Shuan | Suppositories |
| 透皮贴剂 | Tou pi tie ji | Transdermal system |
| 透皮贴片 | Tou pi tie pian | Transdermal patch |
| 外用 | Wai yong | For external use |
| 洗剂 | Xi ji | Wash lotion |
| 洗液 | Xi ye | Wash lotion |
| 注射液 | Zhu she ye | Injection |
| 注射用 | Zhu she yong | For injection |
| 药膏 | Yao gao | Medical ointment |
| 药膏<br>巴布膏 | Yao gao<br>Ba bu gao | Cream<br>Plaster<br>Cataplasm |
| 凝胶贴膏 | Ning jiao tie gao | Cataplasm |
| 膏药 | Gao yao | Plaster with chemical substances |
| 膏 | Gao | Ointment |
| 膏<br>煎膏 | Gao<br>Jian gao | Cream<br>Plaster<br>Decoction |
| 乳膏 | Ru gao | Cream |



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| | | |
|---|---|---|
| 乳剂 | Ru ji | Emulsion |
| 乳液 | Ru ye | Lotion |
| 软膏 | Ruan gao | Ointment |
| 丸 | Wan | Pills |
| 胶丸 | Jiao wan | Soft capsules |
| 肠溶胶丸 | Chang rong jiao wan | Enteric-coated soft capsules |
| 缓释微球 | Huan shi wei qiu | Microspheres sustained release |



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Box 1051, SE-751 40 Uppsala, Sweden
Email: support@who-umc.org, www.who-umc.org